Revolutions Per Kilometer

1 Scope

Note: Nothing in this standard supersedes applicable laws and regulations unless specific exemption has been obtained.

Note: In the event of a conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This standard is written to describe, in general terms, the methods used by General Motors to determine revolutions per kilometer of passenger car, light truck, and compact spare tires using a 1.7 m (67 in) Revolutions Per Kilometer (RPK) test dynamometer under controlled laboratory conditions.

1.2 Foreword. RPK data are used for various purposes including tire validation, odometer calibration, vehicle torque steer modeling and development, analysis of vehicle lead/pull performance, and calibration of anti-locking brake system (ABS) based low tire pressure warning systems as well as others.

1.3 Applicability. The discussion in this standard is restricted to controlled laboratory tests on passenger car, light truck, and compact spare tires. Operating conditions are limited to free rolling tires at dynamic thermal equilibrium with zero slip and inclination angles.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

SAE J2452

2.2 GM Standards/Specifications.

None

2.3 Additional References.

Tire and Rim Association Inc. Yearbook.

Engineering Report TWS080105.

3 Resources

3.1 Facilities. Not Applicable.

3.2 Equipment. Equipment required for RPK testing consists of a 1.7 m (67 in) diameter laboratory test dynamometer, as well as the associated controllers and instrumentation as required for the test method in use. See SAE J2452 for more detail. The SAE J2452 procedure is written for tire Rolling Resistance testing, but RPK measurements are made on similar equipment, hence much of the equipment related discussion applies to RPK as well. Indeed, some test facilities acquire RPK data as part of a rolling resistance test, using the same test equipment.

3.2.1 The roadwheel surface, dimensions, and tolerances must conform to SAE J2452 standards in order to produce RPK data that will correlate well with General Motors RPK measurements.

3.2.2 The test equipment must conform at all times to the machine control and alignment specifications (See Table 1) in order to minimize variation in RPK test data. See Sections 6.4 and 6.6 of SAE J2452 for details.

Table1: Machine Control and Alignment Specs.

PARAMETER	TOLERANCE (±)
Tire Load	the greater of 20 N (5 lbf) or 0.5%
Inflation Pressure	1.5 kPa (0.2 psi)
Test Wheel Speed	1.5 kph (1 mph)
Ambient Temperature	4°C (7°F)
Loaded Radius	1 mm (0.04 in)
Tire Load Fore Aft Offset	0.2 mm (0.01 in)
Tire Load Angular Offset	0.3°
Tire Slip Angle	0.1°
Tire Inclination Angle	0.2°

3.2.3 Machine control accuracy must adhere to standards set in Section 6.6 of SAE J2452. Maintaining these criteria will contribute to precise and unbiased RPK measurements.

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3.2.4 The test drum diameter, without surface treatment, must be 1.708 m (67.23 in).

3.2.4.1 The width of the drum test surface must be sufficient to accommodate the full tread width of the test tire/wheel assembly.

3.2.4.2 The surface of the test drum should be covered with 3M's 80 grade Three-M-ite surface treatment.

3.2.5 Wheel width must be reported with test results. The test wheel must have a contour and width as specified on the work request, and the wheel should be the recommended size for the particular tire to be tested (per the Tire and Rim Association Inc. Yearbook (T&RA) Yearbook). Test wheel runout must meet specifications for new wheels, that is radial runout must be ≤ 0.25 mm (0.01 in) and lateral runout must be ≤ 1.14 mm (0.045 in).

3.2.6 Tire loaded radius values must be measured and recorded with all test data, since these data are later used to estimate flat surface RPK values.

3.3 Test Vehicle/Test Piece.

3.3.1 A minimum sample of five tires per construction should be used to determine RPK_{Bias} and Cpk characteristics.

3.3.2 A minimum sample of two tires per construction should be used to determine RPK magnitudes.

3.4 Test Time. (per tire).

Calendar time:	1 day
Test hours:	2 hours
Coordination hours:	1 hour

3.5 Test Required Information. Exact list of required information is determined by the local test site. Required information includes tires size, construction #, load rating, service description and wheel width.

3.6 Personnel/Skills. Not Applicable.

4 Procedure

4.1 Preparation. Included in instructions in paragraph 4.3.

4.2 Conditions.

4.2.1 Environmental Conditions. The ambient temperature in the test room must be maintained at a steady 24 ± 4°C (75 ± 7°F). Room ambient temperature must be monitored over time to ensure test equipment is at thermal equilibrium before testing takes place.

4.2.2 Test Conditions. Specific test step conditions are described in Table A1. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions/Procedure.

4.3.1 All RPK measurements are made using a test tire mounted on the wheel specified in the work request. The test is conducted on a 1.7 m (67 in) dynamometer which has a test surface covered with a surface treatment in order to provide a first approximation to outdoor public highway surfaces.

4.3.2 Test tires must be mounted on wheels representative of expected usage and whose width and flange type is as specified on the test request.

4.3.3 Tires must be installed on the test wheel such that the cosmetic sidewall of the tire corresponds to the cosmetic face of the test wheel. The cosmetic face can usually be distinguished by the presence of ornamental features such as white lettering, a white stripe, or more ornate sidewall stamping and patterns. The non-cosmetic side usually has more small text, disclaimers, warning labels, and an absence of ornamental features.

4.3.4 Assemblies to be tested must be inflated to the maximum pressure shown on the tire sidewall, then placed in an environment maintained at a constant ambient temperature of 24 ± 4°C (75 ± 7°F) for a minimum of 24 h before a test. This step is required for two reasons: the tire must be allowed to grow to its maximum dimensions, and tire temperature must be close to 24°C (75°F) at the start of the test.

4.3.5 Tires must be tested at test conditions per Table A1, using either the standard tread or directional tread conditions as appropriate to the tread type.

4.3.6 After the tire/wheel assembly is installed on the test machine, the tire must be inflated to the first step point inflation pressure and the tire overall diameter (OD) measured with a Pi tape. If a Pi tape is not used, some other suitably accurate method may be used at another point in the test process to measure tire OD, but the minimum 24 h growth period after initial tire inflation is important and must be maintained.

4.3.7 The initial tire stabilization will take place with the tire rotating in the clockwise direction, and at a speed of 80 kph (50 mph).

4.3.8 The preferred initial tire stabilization time is 40 m, but other times are acceptable if supporting evidence can be provided to indicate the test tire has reached dynamic thermal equilibrium.

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4.3.9 Once the initial stabilization is complete, RPK data is acquired in the clockwise and counter-clockwise direction under the same load, inflation and speed conditions so that RPK_{Bias} may eventually be calculated.

4.3.10 In the nonstandard case that the test has multiple test step conditions for load or pressure, a secondary stabilization period is required before additional RPK data are recorded.

4.3.10.1 Any significant change in load and/or inflation condition will require a stabilization period to allow tire dynamic thermal equilibrium to take place. In the interest of reduced test time, a road wheel direction change or change in speed only does not require a stabilization period. This approach is justified because the thermal effect of speed change on tire RPK is considered to be of second order when compared to the thermal effect of large load and inflation changes.

4.3.10.2 The preferred secondary stabilization time is 10 minutes, but other times are acceptable if supporting evidence can be provided to indicate the test tire has reached dynamic thermal equilibrium.

4.3.11 As a general rule, data for each RPK measurement should be acquired for at least one kilometer. However, this distance may be reduced if it can be demonstrated that test equipment is capable of accuracy to 0.01 RPK or better at the shorter data acquisition distance.

5 Data

5.1 Calculations.

5.1.1 If multiple consecutive RPK_{CW} or RPK_{CCW} measurements are made to improve data accuracy at the same test conditions, the consecutive measurements are to be averaged to determine RPK_{CW} or RPK_{CCW}.

5.1.2 The factor used to convert from roadwheel and test tire encoder counts to RPK must use the actual roadwheel diameter, as measured by a Pi tape (see paragraph 7.1).

5.1.3 Directional bias for individual tires may be determined by subtracting RPK_{CCW} from RPK_{CW}, that is:

$$RPK_{Bias} = RPK_{CW} - RPK_{CCW}$$

5.1.4 Mean directional bias for multiple tires may be determined by averaging all of the RPK_{CCW} values, then all of the RPK_{CW} values, and finally subtracting the average RPK_{CCW} from the average RPK_{CW}, that is:

$$RPK_{AvgBias} = RPK_{AvgCW} - RPK_{AvgCCW}$$

5.1.5 Flat surface RPK values are obtained using the following equation:

$$RPK_F = \frac{RPK_C + \left(\frac{1000}{\pi R}\right)}{1 + \frac{D}{R}}$$

Where:

RPK_F = Flat Surface RPK

RPK_C = Curved Surface RPK

R = Diameter of roadwheel (m)

D = Tire Unloaded OD (m)

Note: Development of this equation is documented in Engineering Report # TWS080105.

5.1.6 Cpk values (see paragraph 7.1) are calculated from a data sample of at least 5 tires using the following equation:

Cpk = (RPKUSL - RPKAvgBias) / (3 * s)

Where:

Cpk = Process capability index

RPKUSL = RPK upper specification limit [rev/km] = 1.0

RPKAvgBias = Mean directional bias [rev/km]

s = Population standard deviation. This value may be estimated by calculating the square root of the sum of the variances of RPKCW and RPKCCW.

5.2. Interpretation of Results. Not Applicable.

5.3 Test Documentation. For test improvement process (TIP) approved data being uploaded to the GM Global Tire Database, follow the electronic file formats in Appendix B. Otherwise, document the test results as shown in the Revs Per Kilometer Summary, sample report, in Appendix C.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.1 Proper procedures and safety equipment in good working order must be in place to ensure operator safety when mounting tires on test wheels, installing tire/wheel assemblies into the test machine, testing the tire/wheel assemblies, and removing tire/wheel assemblies from the test machine.

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6.2 Adequate machine safeguards and proper procedures must be in place to protect personnel working in the area of the test machine in the event of catastrophic tire or wheel failure.

7 Notes

7.1 Glossary.

Applied Load N: The radial force applied to a tire/wheel assembly during a test.

Capped Inflation Pressure kPa: The inflation pressure obtained by inflating the tire, prior to testing, to the required pressure while the tire is at the ambient temperature of the test area, and then sealing the tire with a valve, cap, or other method. As a result of capping, tire inflation pressure will increase during testing as the tire increases in temperature.

Clockwise RPK (RPK_{CW}) *rev/km*: The RPK value associated with a tire rolling on a 1.7 m drum such that the rotation direction corresponds to a tire, cosmetic side out, on the right side of a vehicle moving forward. In other words, an observer viewing the right side of a forward moving vehicle would see clockwise tire rotation.

Counter-clockwise RPK (RPK_{CCW}) *rev/km*: the RPK value associated with a tire rolling on a 1.7 m drum such that the rotation direction corresponds to a tire, cosmetic side out, on the left side of a vehicle moving forward. In other words, an observer viewing the left side of a forward moving vehicle would see counter-clockwise tire rotation.

Cpk: The process capability index, which is a statistical parameter that measures how near a process is running to its specification limits as compared to the natural variability of the process. The larger the index, the less likely it is that any individual item will be outside specification. As a rule of thumb, Cpk must be greater than 1.33 if the process is to be considered capable of consistently producing a part within specification. In the case of RPK, Cpk is used to assess the capability of a supplier to build production tires having acceptable RPK variability. As part of the tire approval process, at least five tires are randomly selected from a production lot and measured to determine production capability.

Directional Tread Tire: A tire whose tread is designed in such a way as to allow rotation in only one direction. This tire type typically has an arrow molded into the sidewall designating the required direction of rotation, and need be tested only in the designated direction.

DOT Serial: A character string, preceded by the abbreviation DOT, of up to 12 alphanumeric characters molded into every tire sidewall during the tire curing process. The serial string is mandated by the United States Department of Transportation and records tire related information such as the facility at which the tire was built, first two characters, the tire size, and the week and year of build, last 3 or 4 characters, as well as other information.

Flat Surface Revolutions Per Kilometer (RPK_F) rev/km: The number of revolutions made by a tire rolling on a flat surface as it traverses a distance of one kilometer under zero slip and inclination angle and fixed load, speed, and regulated inflation

Inflation Pressure kPa: The contained air pressure of an inflated tire.

Pi Tape: A measuring tape, graduated in inches or millimeters multiplied by the factor Pi (3.1416), used to measure tire OD. During measurement, the tape is applied to the tire circumference along the centerline of the tire tread; the tire OD may then be directly read from the tape.

Regulated Inflation Pressure kPa: Regulated inflation pressure involves inflation of the tire to required pressure independent of its temperature, and then maintaining this inflation pressure during testing via an external regulation device. In this case, tire inflation pressure will remain constant during testing regardless of tire temperature.

Revolutions Per Kilometer (RPK) *rev/km (curved)*: The number of revolutions made by a tire rolling on a 1.7 m drum as it traverses a distance of one kilometer under zero slip and inclination angle and fixed load, speed, and regulated inflation pressure.

RPK Directional Bias (RPK_{Bias}) rev/km: The signed difference between RPK_{CW} and RPK_{CCW}, that is:

$$RPK_{Bias} = RPK_{CW} - RPK_{CCW}$$

Speed kph: The tangential velocity of the drum surface on which the tire/wheel assembly rolls.

Standard Tread Tire: A tire whose tread is designed in such a way as to allow installation on either side of the vehicle. Since the tire may be installed to rotate either clockwise or counter-clockwise when the vehicle is moving forward, both RPK_{CW} and RPK_{CCW} data are acquired.

Step Point: The specific set of load, inflation, speed, rotation direction, and time interval operating conditions at which a RPK measurement is to be acquired. The step point values define the detailed test matrix at which the tire is tested.

Test Step: The general set of load and inflation operating conditions at which a RPK measurement is acquired. This parameter is used to characterize the test conditions in general.

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Tire Overall Diameter (OD) mm: The maximum outer diameter, as measured across the highest region of the tread, of an unloaded and inflated tire installed on a wheel.

7.2 Acronyms, Abbreviations, and Symbols.

ABS	Anti-lock Brake System
GMNA	General Motors North America
N	Newton of Force
OD	Overall Diameter
RPK	Revolutions per Kilometer
RPK_{CCW}	Counter-clockwise RPK
RPK_{BIAS}	RPK Directional Bias
RPK_F	Flat Surface RPK
STC	Standard Test Condition
TIP	Test Improvement Process
T&RA	Tire and Rim Association Inc. Yearbook

8 Coding System

This standard shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:

Test to GMW14999

9 Release and Revisions

9.1 Release. This standard originated in January 2006, replacing T92RPK1. It was first approved by Tire Global Subsystem Leadership team in July 2006. It was first published in December 2006.

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Appendix A

RPK Test Conditions

General

- This appendix details the standard test conditions (STC) for the General Motors 1.7 m Revolutions Per Kilometer (RPK) process, as applied to single-point test methodology.
- The Tire and Rim Association Inc. Yearbook (T&RA) referred to in this procedure is an annual publication, and is the industry standard reference for various tire-related parameters as applied to North American vehicle fitment applications.
- Japanese and European passenger car tires are to be tested at the equivalent T&RA P-Metric passenger car tire standard test conditions, i.e. they are to be treated as P-Metric for the purpose of determining test condition parameters.
- Electric vehicle standard test conditions may be determined on a case-by-case basis if not defined in the tables below.
- The terms “Passenger Car Tires” and “Light Truck Tires” refer to the tire type, not the vehicle application. Passenger car tires are those with a “P-Metric” Type designation, and Light Truck Tires are those with a “Light Truck Metric” or “Light Truck High Flotation” designation as standardized by T&RA.

Summary of Conditions

Table A1: RPK Test Conditions

Tire Type	Load Range	Applied Radial Load (1) [N]	Regulated Inflation Pressure [kPa]	Roadwheel Speed [kph]	Test Time Duration [minutes]
Compact Spare “T-Metric”	n/a	80% of Maximum Load Capacity	420	80	40
Passenger Car “P-Metric”	SL	70% of Maximum Load Capacity	228	80	40
Passenger Car “P-Metric”	EL	70% of Maximum Load Capacity	280	80	40
Light Truck “High-Flotation”	ALL	70% of Maximum Load @ 250 kPa	Maximum Tire Pressure	80	40
Light Truck “LT-Metric”	ALL	70% of Max Load @ 350 kPa	Maximum Tire Pressure	80	40

Note: The table in this section describes standard test conditions for the single-point test.
Note[1]: Applied Radial Load is the fixed value as specified, or the indicated percentage of the T&RA value at the specified inflation pressure.

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Appendix B

Electronic File Format Definitions for GM Global Tire Database Submission

Field No.	Variable	Field Format	Example	Comments
GENERAL INFORMATION				
1	Tire Test Request No.	A8	99-135	Supplied by GM
2	Test Site	A3	xxx	Valid codes only (see GM Global Tire Database)
3	Tester	A3	xxx	Valid codes only (see GM Global Tire Database)
4	End Test Date	A10	07/19/1998	[mm/dd/yyyy] Date of final test point completion of final tire
5	Comment	A30	Development	Indicate if data are from Development, Initial Production Sample (IPS), Regular Production Tire, or Correlation Tire.
COMMOM TIRE TEST INFORMATION				
6	Size	A13	P235/60R16	Tire size as shown on sidewall
7	Load Rating	A2	SL	Valid codes only (see GM Global Tire Database)
8	Brand	A3	xxx	Valid codes only (see GM Global Tire Database)
9	Construction No.	A12	123456	Character string to uniquely identify a tire construction
10	Rim Width	A5	7.00 or 225	[in] or [mm], as necessary to define rim width
11	Inflation Pressure	I3	260	[kPa] Inflation pressure
12	Test Load	I5	5320	[N] Test load
13	Test Speed	I3	80	[kph] Test speed
14	Warm-up Period	I3	40	[min] Initial tire warm-up time
15	Number of Tires	I3	2	Must equal number of data sets below
16	Tread Type	A3	STD	[STD] or [DIR] only, Standard tread or Directional tread
INDIVIDUAL TIRE TEST DATA				
17	Tire ID1	A10	GM-105	Character string to uniquely identify a specific tire within a tire construction
18	DOT Serial1	A6	xx1797	First 2 + last 3 characters, or first 2 + last 4 characters if necessary to accommodate 2-digit year
19	Test Machine & Station1	A6	Sta. 2	Test machine and station on which data are acquired
20	Test Date1	A10	07/19/1998	[mm/dd/yyyy] Date of final test point completion of this tire
21	CW RPK1	F6.2	511.89	[rev/km] Clockwise Revolutions per Kilometer
22	CCW RPK1	F6.2	511.78	[rev/km] Counterclockwise Revolutions per Kilometer

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23	Tire OD1	F6.2	647.33	[mm] Overall tire diameter
24	Tire ID2	A10	GM-106	Character string to uniquely identify a specific tire within a tire construction
25	DOT Serial2	A6	xx1797	First 2 + last 3 characters, or first 2 + last 4 characters if necessary to accommodate 2-digit year
26	Test Machine & Station2	A3	S2	Test machine and station on which data are acquired
27	Test Date2	A10	07/19/1998	[mm/dd/yyyy] Month/Day/Year of final test point completion of this tire
28	CW RPK2	F6.2	511.95	[rev/km] Clockwise Revolutions per Kilometer
29	CCW RPK2	F6.2	511.83	[rev/km] Counterclockwise Revolutions per Kilometer
30	Tire OD2	F6.2	647.24	[mm] Overall tire diameter
31	⋮			

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Appendix C

Revs Per Kilometer Summary (Sample Only)

SIZE: P275/55R20 LR: SL LOAD INDEX: 111 SPEED RATING: S TYPE: PURPS

SUPPLIER: XXX RIM (IN): 8.5 INF (kPa): 207 LOAD (N): 7919

												<---1.7 METER RPK--->				<--PRED. FLAT RPK-->			
PDF WORK REQ	**CONSTRUCTION**	**C**	**S**	**OBJ**	**BUILD**	**DATE**	**TIRE ID**	**REQ**	**LOC**	**TST**	**OD (MM)**	**CW**	**CCW**	**DIFF**	**AVG**	**CW**	**CCW**	**DIFF**	**AVG**
06-001	XX0XYZ	S	S	REL	XY000Z	02/09/2004	2	PM	TWS	TWS	813.2	405.51	405.51	0.01	405.51	400.97	400.97	0	400.97
06-001	XX0XYZ	S	S	REL	XY000Z	02/09/2004	4	PM	TWS	TWS	812.7	405.77	405.74	0.03	405.76	401.23	401.21	0.02	401.22
06-001	XX0XYZ	S	S	REL	XY000Z	02/09/2004	3	PM	TWS	TWS	813.5	405.38	405.38	0	405.38	400.84	400.84	0	400.84
06-001	XX0XYZ	S	S	REL	XY000Z	02/09/2004	1	PM	TWS	TWS	812.3	406.16	406.12	0.04	406.14	401.56	401.53	0.03	401.55
06-001	XX0XYZ	S	S	REL	XY0001	01/08/2004	0058	PM	XYZ	XXX	812.29	404.99	404.92	0.07	404.96	400.77	400.72	0.05	400.75
06-001	XX0XYZ	S	S	REL	XY0001	01/08/2004	0059	PM	XYZ	XXX	812.29	404.84	404.78	0.06	404.81	400.66	400.62	0.04	400.64
06-001	XX0XYZ	S	S	REL	XY0001	01/08/2004	0060	PM	XYZ	XXX	812.29	404.81	404.75	0.06	404.78	400.64	400.6	0.04	400.62
06-001	XX0XYZ	S	S	REL	XY0001	01/08/2004	0061	PM	XYZ	XXX	812.29	405.22	405.2	0.02	405.21	400.92	400.91	0.01	400.92
							AVERAGE FOR N = 8				812.61	405.34	405.3	0.04	405.32	400.95	400.93	0.02	400.94
							STANDARD DEV.				0.486	0.471	0.484	0.026	0.478	0.311	0.316	0.019	0.312

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